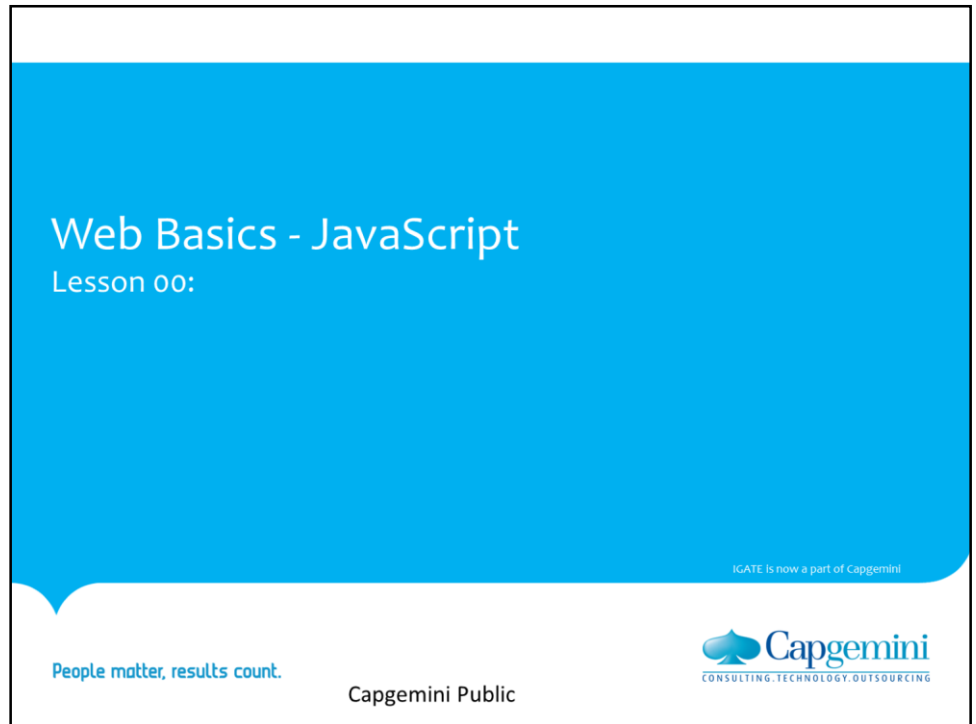


Instructor Notes:



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Instructor Notes:

Document History

Date	Course Version No.	Software Version No.	Developer / SME	Change Record Remarks
19-Sep-2009	2.0	NA	Pradnya Jagtap	Updated to new template. And incorporated new examples.
Apr-2011	3.0	NA	Anu Mitra	Integration updates
Apr-2015	3.1	NA	Rathnajothi P	Revamped according to revised curriculum
May-2016	3.2	NA	Kavita Arora	Revamped according to revised curriculum

Instructor Notes:

Course Goals and Non Goals

➤ Course Goals:

- At the end of this course you will be able to:
 - Add interactivity to the static HTML pages
 - Validate the input data provided by users, on the client side
 - Manipulate style sheets on the fly to give a sophisticated look to any website



➤ Course Non Goals:

- Server-side scripting not covered.

Instructor Notes:

Pre-requisites

➤ **Prerequisites for this course are:**

- Familiarity with Windows, GUI Concept and Web Browser Application
- HTML
- Experience of creation of a web page using HTML
- Object Oriented Programming Concepts

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Instructor Notes:

Intended Audience

- **This course is designed for:**
 - Trainee Programmers
 - Software Professionals



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Instructor Notes:

Day Wise Schedule

➤ **Day 1**

- Lesson 1: Introduction to JavaScript
- Lesson 2: JavaScript Language

➤ **Day 2**

- Lesson 3: Working with Predefined Core Objects
- Lesson 4: Working with Arrays
- Lesson 5: Document Object Model
- Lesson 6: Working with Document object
- Lesson 7: Working with Form object

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Instructor Notes:

Table of Contents

- **Lesson 1: Introduction to JavaScript**
 - 1.1. Basic Concepts of JavaScript
 - 1.2. Embedding JavaScript in HTML
 - 1.3 Writing JavaScript
- **Lesson 2: JavaScript Language**
 - 2.1. Data Types and Variables
 - 2.2. JavaScript Operators
 - 2.3. Control Structures and Loops
 - 2.4. JavaScript Functions
- **Lesson 3: Working with Predefined Core Objects**
 - 3.1. Data Types in JavaScript
 - 3.2. Overview of String object
 - 3.3. Math object
 - 3.4 Date object

Instructor Notes:

Table of Contents

- **Lesson 4: Working with Arrays**
 - 4.1. Array Object
 - 4.2. Properties and Methods of Array object
- **Lesson 5: Document Object Model**
 - 5.1. JavaScript Document Object Model
 - 5.2. Window object
 - 5.3. Navigator Object
 - 5.4. Location Object
 - 5.5. History Object
- **Lesson 6: Working With Document Object**
 - 6.1. Document Objects
 - 6.2. Cookies

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Instructor Notes:

Table of Contents

➤ **Lesson 7: Working with Form Object**

- 7.1. Form properties, Methods & Event handlers
- 7.2. Text-Related Objects
- 7.3. Button Objects
- 7.4. Check Box and Radio Objects
- 7.5. Select Objects

Instructor Notes:

References

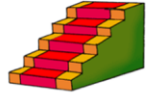
- JavaScript: A Beginner's Guide - by John Pollock
- <http://www.w3schools.com/>



Instructor Notes:

Next Step Courses (if applicable)

- Servlet
- JSP



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Instructor Notes:

Other Parallel Technology Areas

- VBScript

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Instructor Notes:



Web basics-JavaScript

Lesson 1: Introduction to JavaScript

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Instructor Notes:

Lesson Objectives

➤ **To understand the following topics:**

- Basic Concepts of JavaScript
- Embedding JavaScript in HTML
- Writing JavaScript

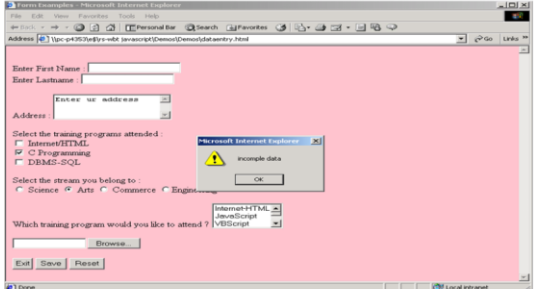


Instructor Notes:

1.1: Basic Concepts of JavaScript

Basic Concepts of JavaScript

- JavaScript is the scripting language of the Web
- JavaScript is used for placing dynamic content into HTML page and for client side validation which cannot be performed with HTML 5 elements and attributes



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Basic Concepts of JavaScript

JavaScript History :

Web pages made using only HTML are somewhat static with no interactivity and negligible user involvement.

HTML tags are just instructions on document and the display of the document is dependent on the browser.

Interactive pages cannot be built with only HTML, we need a programming language.

So Netscape came out with a client-side language called as JavaScript.

What is JavaScript?

JavaScript is THE scripting language of the Web.

JavaScript is used in Web pages to add functionality, validate forms, detect browsers, and much more.

JavaScript is the most popular scripting language on the internet and works on all major browsers available like IE, Firefox, Chrome etc..

Need of JavaScript:

For placing dynamic content into an HTML Page.

For client side Validation.

For storing and retrieving client's information in the form of Cookies.

Instructor Notes:

1.1: Basic Concepts of JavaScript

Overview

- JavaScript is Netscape's cross-platform, object-based scripting language
- JavaScript code is embedded into HTML pages
- It is a lightweight programming language
- Client-side JavaScript extends the core language by supplying objects to control a browser and its Document Object Model
- Server-side JavaScript extends the core language by supplying objects relevant to running JavaScript on a server

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- 16 -

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What is JavaScript?

JavaScript is Netscape's cross-platform, object-based scripting language. Core JavaScript contains a core set of objects, such as Array, Date, and Math, and a core set of language elements such as operators, control structures, and statements. Core JavaScript can be extended for a variety of purposes by supplementing it with additional objects; for example:

Client-side JavaScript extends the core language by supplying objects to control a browser (Navigator or another web browser) and its Document Object Model (DOM). For example, client-side extensions allow an application to place elements on an HTML form and respond to user events such as mouse clicks, form input, and page navigation.

Server-side JavaScript extends the core language by supplying objects relevant to running JavaScript on a server. For example, server-side extensions allow an application to communicate with a relational database, provide continuity of information from one invocation to another of the application, or perform file manipulations on a server.

JavaScript lets you create applications that run over the Internet. Client applications run in a browser, such as Internet Explorer/Firefox, and server applications run on a server, such as Netscape Enterprise Server. Using JavaScript, you can create dynamic HTML pages that process user input and maintain persistent data using special objects, files, and relational databases.

Instructor Notes:

1.1: Basic Concepts of JavaScript

How does it work ?

➤

When JavaScript code is inserted into an HTML document, and the HTML document is opened in a web browser:

—

The browser will read the HTML .

—

The JavaScript interpreter which is built-in within the browser interprets the JavaScript.

—

It executes the JavaScript immediately, or at a later event i.e could be based on user action or system event.

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- 17 -

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How does JavaScript work?

When a JavaScript is inserted into an HTML document, the JavaScript interpreter which is built-in within the Internet browser will read the HTML and interpret the JavaScript. The JavaScript can be executed immediately, or at a later event.

Instructor Notes:

1.1: Basic Concepts of JavaScript

Why use JavaScript?

➤

JavaScript:

- Provides HTML designers a programming tool
- Places dynamic text into an HTML page
- Reacts to events
- Reads and writes to HTML elements
- Can be used to validate form data.
 - for validation not possible using HTML 5 elements and attributes

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- 18 -

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Why use JavaScript?

JavaScript gives HTML designers a programming tool :

JavaScript is a simple scripting language which can be used by HTML authors who essentially are not familiar with programming. Hence the HTML authors can easily put small JavaScript code snippets into HTML pages.

JavaScript can place dynamic text into an HTML page : A simple HTML text which displays static content such as `<h1>Welcome</h1>` can be written in JavaScript to display dynamic content using `document.write("<h1>" + orname + "</h1>")`

JavaScript can react to events : A JavaScript can be set to execute when some action takes place, like when a page has finished loading or when a user clicks on an HTML element.

JavaScript can reads and writes HTML elements: A JavaScript can read values of HTML elements and also write/change the content of an HTML element.

JavaScript can be used to validate data : Client side validation can easily taken care of by JavaScript. This reduces the burden on the server.

With the advent of HTML 5, most of the form data validation can be easily performed using HTML 5 elements like `input type=email` and attributes like `required`, `pattern`, `maxlength` etc

Apart from this JavaScript can also be used to create cookies which is stores and retrieves information about the user preferences. JavaScript can also be used to detect browser which helps in loading a page specifically designed for the browser.

Instructor Notes:

1.2: Embedding JavaScript in HTML

Embedding JavaScript in HTML

➤ The <SCRIPT> tag

```
<SCRIPT>
  JavaScript statements ...
</SCRIPT>
```

➤ Specifying the JavaScript version

```
<SCRIPT LANGUAGE="JavaScript 1.2">
```

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- 19 -

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Embedding JavaScript in HTML:

The <SCRIPT> tag is an extension to HTML that can enclose any number of JavaScript statements as shown on the slide.

A document can have multiple <SCRIPT> tags, and each can enclose any number of JavaScript statements.

The Script tag has the following attributes:

Language – This attribute specifies the scripting language. It can have values like VBScript, JavaScript. Optionally you can also specify the scripting language version as mentioned on the slide.

Type – Specifies the MIME type of the scripting language

Src – This attribute is for specifying the path and filename of an external .js file which contains the script code.

Some browsers may not support JavaScript and the JavaScript code is displayed as page content. To prevent this, the HTML comment should be used which will hide the JavaScript as shown on the slide. Note the two forward slashes at the end of the comment line. This is JavaScript comment symbol which prevents JavaScript from executing the --> tag.

Note that Javascript is a case-sensitive scripting language.

Instructor Notes:

1.2: Embedding JavaScript in HTML

Embedding JavaScript in HTML(contd)

➤

Hiding Scripts with Comment tags

```
<script type="text/javascript">  
    <!--  
        some statements ...  
    // -->  
</script>
```

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- 20 -

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Instructor Notes:

1.2: Embedding JavaScript in HTML

Embedding JavaScript in HTML (Contd.)

- Using Quotation Marks

```
document.write("<A HREF='A.HTML'>Link to next page")
```
- Specifying alternate content with the NOSCRIPT tag

```
<NOSCRIPT>  
    Your browser has JavaScript turned off.  
</NOSCRIPT>
```

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Embedding JavaScript in HTML (contd.):

Whenever you want to indicate a quoted string inside a string literal, use single quotation marks (') to delimit the string literal. This allows the script to distinguish the literal inside the string. In the following example, the attribute values are in double quotes, but in the call to the function `myfunc` the argument passed is a string which is enclosed in a single quote.

```
<INPUT TYPE="button" VALUE="Press Me" onClick="myfunc('astring')">.
```

Use the `<NOSCRIPT>` tag to specify alternate content for browsers that do not support JavaScript. HTML enclosed within a `<NOSCRIPT>` tag is displayed by browsers that do not support JavaScript; code within the tag is ignored by browser. In case, if the user has disabled JavaScript from the Advanced tab of the Preferences dialog, the browser displays the code within the `<NOSCRIPT>` tag.

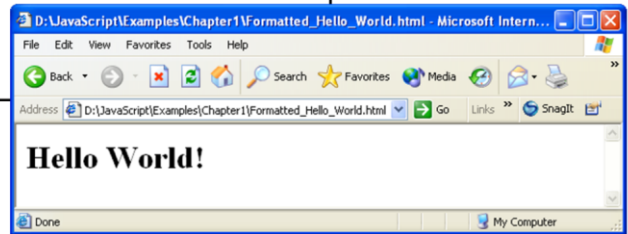
Instructor Notes:

1.2: Embedding JavaScript in HTML

Embedding JavaScript in HTML (Contd.)

➤ Including text-formatting features

```
<!DOCTYPE html>
<html>
<head> </head>
<body>
<script>
document.write("<H1>Hello World!</H1>")
</script>
</body>
</html>
```

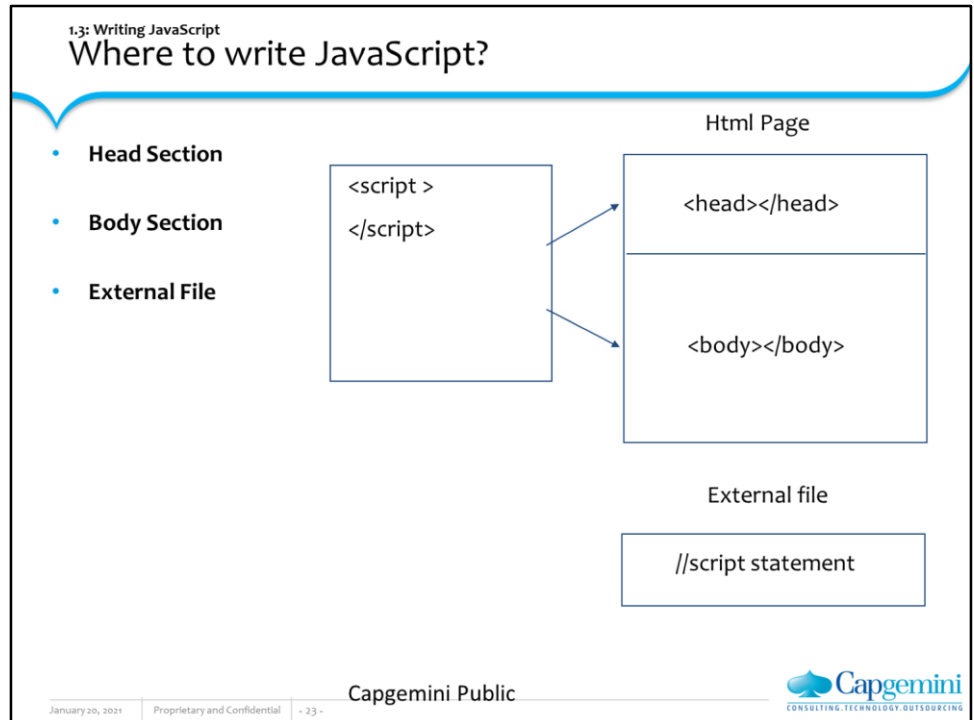


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Instructor Notes:



Embedding JavaScript in HTML: Where to Write JavaScript?

Physically the script code can be placed at three different places in the html file.

Head Section: If you want the Scripts to be executed when they are called, or whenever a user action happens, put script tag in the head section.

Body Section : If you want the Scripts to be executed when the page loads then put script tag in the body section

External File : If you want to run the same JavaScript on several pages, without having to write the same script on every page, you can write a JavaScript in an external file.

Instructor Notes:

1.3: Writing JavaScript

JavaScript in Head Section

➤ Syntax

```
<!DOCTYPE html>
<html>
<head>
<script type="text/javascript">
function message()
{
    alert("This alert box was called with the
        onload event")
}
</script>
</head>
<body onload="message()">
</body>
</html>
```

January 20, 2001 Proprietary and Confidential - 24 -

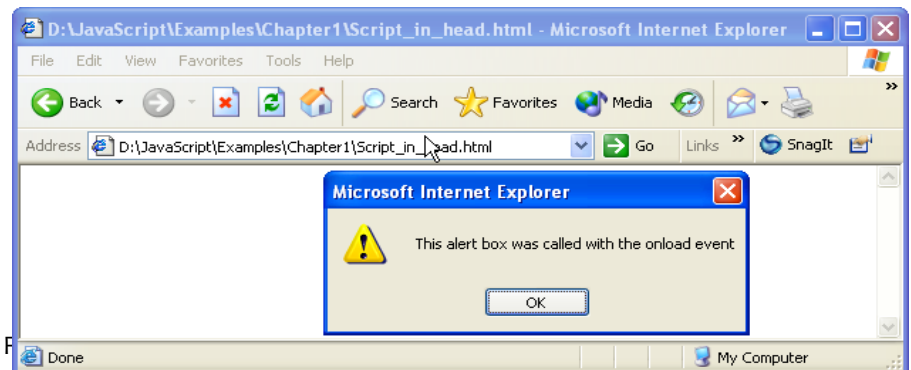
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Where to Write JavaScript? In the Head Section:

Head Section : JavaScripts in an HTML page will be executed when the page loads. This might always not be the case. Sometimes we want to execute a JavaScript when an event occurs, such as when a user clicks a button. In such scenarios, we can put the script inside a function. Scripts that contain functions go in the head section of the document. Then we can be sure that the script is loaded before the function is called.

The example shown on the slide shows the following output:



Instructor Notes:

1.3: Writing JavaScript
JavaScript in Body Section

➤ **Syntax**

```
<!DOCTYPE html>
<html>
<head>
<title>script tag in body</title>
</head>
<body >
<script language="javascript">
    document.write("this message is written when
the page loads")
</script>
</body>
</html>
```

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Where to Write JavaScript? In the Body Section:

Body Section : Execute a script that is placed in the body section. The example on the slide shows script written in the body section.

And it produces this output:

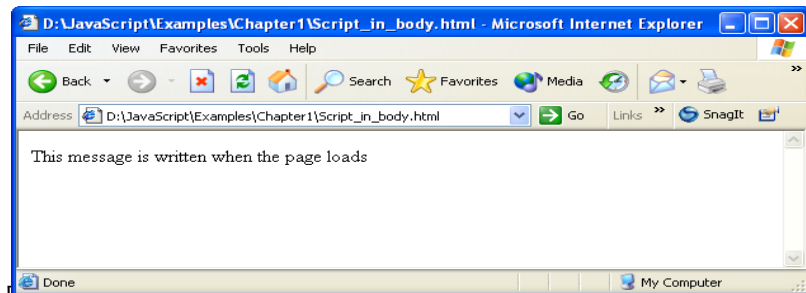


Figure 1.4 Output of Example 1.4 (Script_in_body.html).

Instructor Notes:

1.3: Writing JavaScript

JavaScript in External File

- **Content of common.js**

```
var msg  
msg="<h1>declared in external js file</h1>"
```
- **HTML page using the external script file**

```
<!DOCTYPE html>  
<html>  
<head><title>script tag in external file</title>  
<script src="common.js">  
<!-- no javascript statements can be written here-->  
</script>  
</head>  
<body> <script>  
document.write("display value of a variable"+msg)  
</script> </body>  
</html>
```

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- 26 -


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Where to Write JavaScript? In External File:

The example on the slide demonstrates how to write JavaScript code in an external file. The extension of the external file has to be .js and the it does not contain the script tag.

Instructor Notes:

Demo

- Hello.html
- Head_section.html
- Extern_file.html
- Comm.js
- Var_ex.html



Instructor Notes:

Lab

- **Lab 1:**
- **Basic concepts of JavaScript**



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- 28 -

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Instructor Notes:

Summary

- JavaScript is the client side scripting language
- When JavaScript code is placed inside a web page, the browser loads the page & the built-in interpreter reads the script & executes the same
- JavaScript is used in Web pages for
 - Validating client side data.
 - Placing dynamic content into an HTML page.
 - Storing client's information in the form of Cookies.



Instructor Notes:

Answers

1. Option 2
2. False
3. noscript

Review Question

➤ **Question 1: JavaScript, a scripting language, is:**

- Option 1: Cross-platform, object-oriented
- Option 2: Cross-platform, object-based
- Option 3: Non cross-platform, object-oriented



➤ **Question 2: The <SCR> tag is an extension to HTML that can enclose any number of JavaScript statements.**

- True/False

➤ **Question 3: HTML enclosed within a _____ tag is displayed by browsers that do not support JavaScript.**

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- 30 -

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Instructor Notes:

Answers

- 1. 3
- 2. 1
- 3. 4
- 4. 2

Review Question: Match the Following

1. Embed JavaScript in an HTML document as statements and functions within this tag

2. To specify alternate content for browsers that do not support JavaScript

3. Feature of JavaScript

4. JavaScript can be written in these sections

1. <NOSCRIPT>

2. Head, Body

3. SCRIPT

4. To validate and process user data



Instructor Notes:



Web basics-JavaScript

Lesson 2: JavaScript Language

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Instructor Notes:

Lesson Objectives

- Data Types and Variables
- JavaScript Operators
- Control Structures and Loops
- JavaScript Functions



Instructor Notes:

2.1: Data Types and Variables

Data Types in JavaScript

- **JavaScript is a free-form language. You do not have to declare all variables, classes, and methods**
- **Variables in JavaScript can be of type:**
 - Number (4.156, 39)
 - String ("This is JavaScript")
 - Boolean (true or false)
 - Null (null)

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- 34 -

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Data Types in JavaScript:

Although the number of data types is small, they are sufficient for the tasks that JavaScript performs. Notice that there is no distinction between integers and real numbers; both types are just numbers. JavaScript does not provide an explicit data type for a date. However, there are related functions and a built-in date object that enable the Web page designer to manage dates.

Instructor Notes:

2.1: Data Types and Variables

Data Types in JavaScript (Contd..)

- **JavaScript variables are said to be loosely typed**
- **Defining variables:** `var variableName = value`
- **JavaScript variables**
 - Can include letters of the alphabet, digits 0-9 and the underscore (`_`) character and is case-sensitive.
 - Cannot include spaces or any other punctuation characters.
 - First character of the variable name must be either a letter or the underscore character.
 - No official limit on the length of a variable name, but must fit within a line.

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**Defining Variables:**

There are specific rules you must follow when choosing variable names.

Variable names can include letters of the alphabet, both upper and lowercase.

They can also include the digits 0-9 and the underscore (`_`) character.

Variable names cannot include spaces or any other punctuation characters.

The first character of the variable name must be either a letter or the underscore character.

Variable names are case-sensitive; `totalnum`, `Totalnum`, and `TotalNum` are separate variable names.

There is no official limit on the length of a variable name, but it must fit within one line.

JavaScript variables are said to be loosely typed. To declare a variable for a JavaScript program, you would write this:

```
var variablename = value;
```

Instructor Notes:

2.2: JavaScript Operators

Arithmetic Operator

Operator	Description	Example	Result
+	Addition	2 + 2	4
-	Subtraction	5 - 2	3
*	Multiplication	4 * 5	20
/	Division	5 / 2	2.5
%	Modulus	10 % 8	2
++	Increment	x = 5; x++	x = 6
--	Decrement	x = 5; x--	x = 4

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- 36 -

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Instructor Notes:

2.2: JavaScript Operators

Comparison Operator

Operator	Description	Example	Result
==	is equal to	5 == 8	false
!=	is not equal	5 != 8	true
>	is greater than	5 > 8	false
<	is less than	5 < 8	true
>=	is greater or equal	5 >= 8	false
<=	is less or equal	5 <= 8	true

Instructor Notes:

2.2: JavaScript Operators

Assignment Operator

Operator	Example	Is same as
<code>+=</code>	<code>x += y</code>	<code>x = x + y</code>
<code>-=</code>	<code>x -= y</code>	<code>x = x - y</code>
<code>*=</code>	<code>x *= y</code>	<code>x = x * y</code>
<code>/=</code>	<code>x /= y</code>	<code>x = x / y</code>
<code>%=</code>	<code>x %= y</code>	<code>x = x % y</code>

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- 38 -

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Instructor Notes:

2.2: JavaScript Operators

Logical Operator

Operator	Description	Example
&&	and	x = 6; y = 3 x < 10 && y > 1 returns true
	or	x = 6; y = 3 x < 10 y > 5 returns true
!	not	x = false !x returns true

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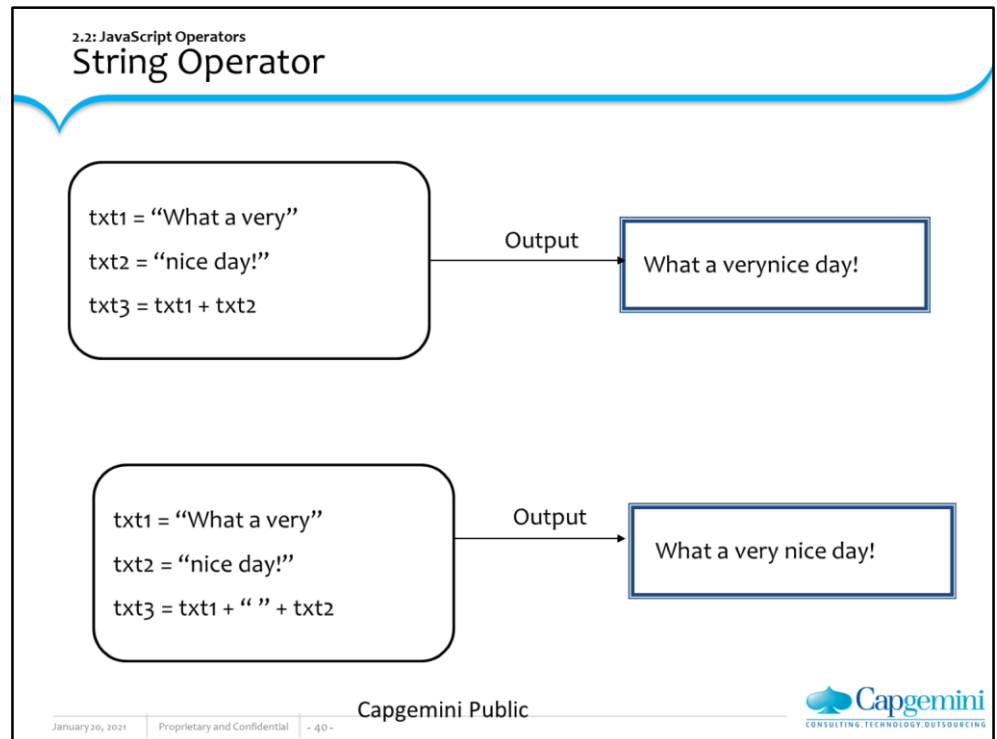
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- 39 -

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Instructor Notes:

**String Operator:**

```
txt1="What a very"
txt2="nice day!"
txt3=txt1+txt2
```

A string is most often a text, for example "Hello World!". To stick two or more string variables together, use the + operator.

The variable txt3 now contains "What a verynice day!". To add a space between two string variables, insert a space into the expression, OR in one of the strings.

```
txt1="What a very"
txt2="nice day!"
txt3=txt1+" "+txt2
```

Or

```
txt1="What a very "
txt2="nice day!"
txt3=txt1+txt2
```

The variable txt3 now contains "What a very nice day!".

Instructor Notes:

2.2: JavaScript Operators Typeof Operator		
typeof	undefinedvariable	"undefined"
typeof	33	"number"
typeof	"abcdef"	"string"
typeof	true	"boolean"
typeof	null	"object"

January 30, 2021 Proprietary and Confidential - 41 -

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Typeof Operator:

The typeof operator returns the type of data that its operand currently holds.

```
<HTML>
<HEAD>
<TITLE>Using typeof</TITLE>
</HEAD>
<BODY>
<SCRIPT LANGUAGE="JavaScript">
<!-- var num1=20
var str1="abc"
var bool1=true
var num2=null
var var1;
document.write("type of str1 : "+typeof(str1)+"<BR>")
document.write("type of num1 : "+typeof(num1)+"<BR>")
document.write("type of bool1 : "+typeof(bool1)+"<BR>")
document.write("type of num2 : "+typeof(num2)+"<BR>")-->
</SCRIPT>
</BODY>
</HTML>
```

Example 2.1 typeof operator (typeof_ex.html)

Instructor Notes:

Demo

➤ `Typeof_ex.html`



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- 42 -

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Instructor Notes:

2.3: Control Structures and Loops

Control Structures and Loops

➤ JavaScript supports the usual control structures:

— the conditionals:

- if,
- if...else
- If ... else if ... else
- switch

— iterations:

- for
- while

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- 43 -

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Control Structures and Loops

Conditional statements are used to perform different actions based on different conditions. Loops execute a block of code for specified number of times or while a specified condition is true.

Instructor Notes:

2.3: Control Structures and Loops
The if Statement

```
if(condition){  
    statement 1  
} else {  
    statement 2  
}
```

```
if(a>10) {  
    document.write("Greater than 10")  
} else {  
    document.write("Less than 10")  
}
```

Shorthand

```
document.write( (a>10) ? "Greater than 10" : "Less than 10" );
```

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The if Statement:

The condition is any JavaScript expression that evaluates to the Boolean type, either true or false.

The example as shown on the slide.

A shorthand method can also be used for these types of statements, where ? indicates the 'if' portion and : indicates the 'else' portion. This statement is equivalent to the previous example:

The equivalent shorthand method is also seen on the slide.

Instructor Notes:

2.3: Control Structures and Loops

The Switch Statement

>

Syntax

```
switch (variable) {  
  case outcome1 : {  
    //stmts for outcome 1  
    break; }  
  case outcome2 : {  
    //stmts outcome 2  
    break; }  
  ...  
  default: {  
    //none of the outcomes  
    is chosen }  
}
```

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- 45 -


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Instructor Notes:

2.3: Control Structures and Loops

The Switch Statement

➤ Code Snippet

```
switch (day){  
  case "Monday": {  
    document.write("weekday")  
    break;}  
  case "Saturday": {  
    document.write("weekday")  
    break}  
  ...  
  default: {  
    document.write("Invalid day of the week")  
  }  
}
```

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- 46 -

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Instructor Notes:

2.3: Control Structures and Loops

The for Statement

➤ **Syntax**

```
for( [initial expression;][condition;][increment expression] )
{
    statements
}
```

➤ **Code Snippet**

```
for(var i=0;i<10;i++){
    document.write("Hello");}
```

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Looping Statements

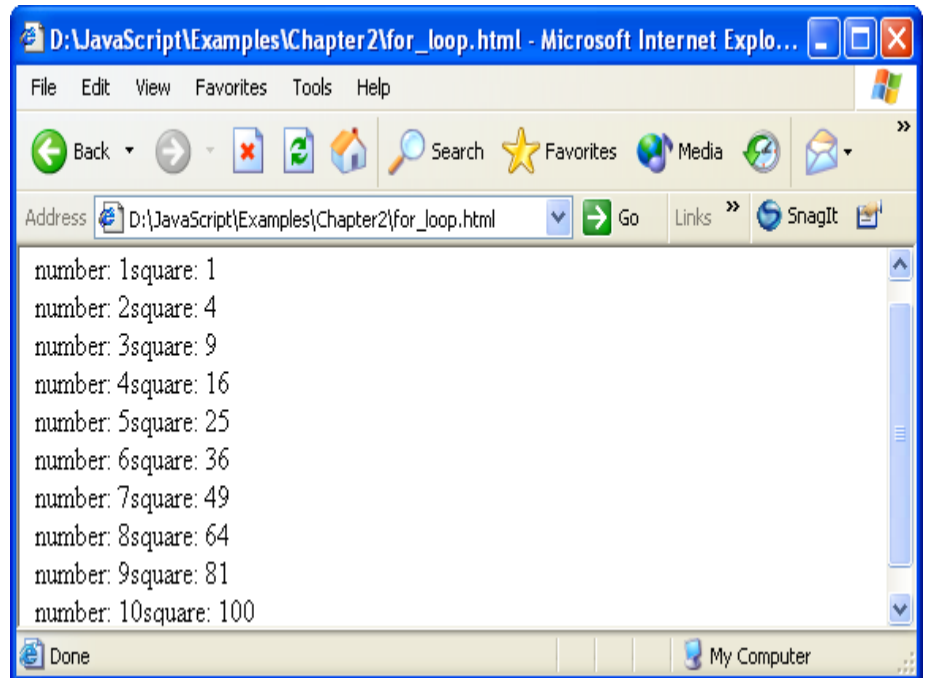
The 'for' Statement:

```
<HTML>
<HEAD>
<SCRIPT LANGUAGE="JavaScript">
<!-- hide script
for (i=1; i<=10; i++)
{
    sq=i*i
    document.write("number: " + i + "square: " + sq + "<BR>")
}
// end script hiding -->
</SCRIPT>
</HEAD>
<BODY></BODY></HTML>
```

Example 2.2 For Construct (for_ex.html)

And it produces the output as:

Instructor Notes:

**The 'while' Statement**

The while statement continues to repeat the loop as long as the condition is true. The syntax for the while statement is as follows:
while (condition):

```
{  
    statements  
}
```

```
<HTML>  
<HEAD>  
<SCRIPT LANGUAGE="JavaScript">  
<!-- hide script  
i=1  
while (i<=10)  
{  
    sq=i*i  
    document.write("number: " + i + "square: " + sq + "<BR>")  
    i++  
}  
// end script hiding -->  
</SCRIPT>  
</HEAD>  
<BODY></BODY>  
</HTML>
```

Example 2.3 While Construct
And it produces the output as in the previous screen shot.

Instructor Notes:

2.3: Control Structures and Loops

The break and continue Statements

➤ break

– Writing break inside a switch, for, while control structure will cause the program to jump to the end of the block. Control resumes after the block, as if the block had finished

➤ continue

– Writing continue inside a loop will cause the program to jump to the test condition of the structure and re-evaluate and perform instruction of the loop. Control resumes at the next iteration of the loop

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- 49 -

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The break statement:

This statement is used to break out of the current 'for' or 'while' loop. Control resumes after the loop, as if it had finished.

The continue Statement:

This statement continues a 'for' or 'while' loop without executing the rest of the loop. Control resumes at the next iteration of the loop.

Instructor Notes:

Demo

➤ For_ex.html



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- 50 -

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Instructor Notes:

2.4: JavaScript Functions

JavaScript Functions

The function statement

```
function myFunction (arg1, arg2, arg3)
{
    statements
    return //The return keyword returns a value.
}
```

How to call a function

```
myFunction( "abc", "xyz", 4 )
or
myFunction()
```

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The function Statement:

A function contains some code that will be executed by an event or a call to that function. A function is a set of statements. You can reuse functions within the same script, or in other documents. You define functions at the beginning of a file (in the head section), and call them later in the document.

The syntax of a typical function is as follows:

```
function myfunction(arg1, arg2, arg3)
{
    statements
}
```

How to Call a Function?

A function is not executed before it is called. You can call a function containing arguments:

```
myfunction("abc","xyz",4)
or without arguments:
myfunction()
```

Instructor Notes:

2.4: JavaScript Functions

Argument Arrays and How to call a Function

- Syntax for the arguments array:

```
arguments[index]  
functionName.arguments[index]
```

index – ordinal number of the argument starting at zero
arguments.length – Total number of arguments

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- 52 -

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Using the arguments Array

The arguments of a function are maintained in an array. Within a function, you can address the parameters passed to it as follows:

`arguments[i]`

`functionName.arguments[i]`

where *i* is the ordinal number of the argument, starting at zero. So, the first argument passed to a function would be `arguments[0]`. The total number of arguments is indicated by `arguments.length`. Using the arguments array, you can call a function with more arguments than it is formally declared to accept. This is often useful if you don't know in advance how many arguments will be passed to the function. You can use `arguments.length` to determine the number of arguments actually passed to the function, and then treat each argument using the arguments array.

Instructor Notes:

The Function Statement (Contd..)

➤ Syntax

```
function myConcat(separator) {
    result = ""
    for(var index=1; index<arguments.length;index++) {
        result += arguments[index] + separator
    }
    return result
}
```

➤ Code Snippet

```
myConcat( ",", "red", "orange", "blue")
// returns "red, orange, blue"
```

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- 53 -

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For example, consider a function that concatenates several strings. The only formal argument for the function is a string that specifies the characters that separate the items to concatenate. The function is defined as shown on the slide.

You can pass any number of arguments to this function, and it creates a list using each argument as an item in the list.

```
// returns "red, orange, blue, "
myConcat( ",", "red", "orange", "blue")
// returns "elephant; giraffe; lion; cheetah;"
myConcat( ";", "elephant", "giraffe", "lion", "cheetah")
// returns "sage. basil. oregano. pepper. parsley."
myConcat( ".", "sage", "basil", "oregano", "pepper", "parsley")
```

Instructor Notes:

2.4: JavaScript Functions

Predefined Functions

- eval:**
 - Evaluates a string of JavaScript code without reference to a particular object.

`eval (expr)`
 where expr is a string to be evaluated
- isFinite:**
 - Evaluates an argument to determine whether it is a finite number.

`isFinite (number)`
 where number is the number to evaluate

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- 54 -

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Predefined Functions:

JavaScript has several top-level predefined functions:

Eval, isFinite, isNaN, parseInt and parseFloat, Number and String

eval Function

The eval function evaluates a string of JavaScript code without reference to a particular object. The syntax of eval is:

`eval(expr)` where expr is a string to be evaluated.

If the string represents an expression, eval evaluates the expression. If the argument represents one or more JavaScript statements, eval performs the statements. Do not call eval to evaluate an arithmetic expression; JavaScript evaluates arithmetic expressions automatically.

isFinite Function

The isFinite function evaluates an argument to determine whether it is a finite number. The syntax of isFinite is:

`isFinite(number)` where number is the number to evaluate.

If the argument is NaN, positive infinity or negative infinity, this method returns false, otherwise it returns true. The following code checks client input to determine whether it is a finite number.

```
if(isFinite(ClientInput)== true)
{
    /* take specific steps */
}
```

Instructor Notes:

2.4: JavaScript Functions

Predefined Functions (Contd..)

➤ **isNaN :**

- Evaluates an argument to determine if it is “NaN” (not a number)

`isNaN (testValue)`
where testValue is the value you want to evaluate

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isNaN Functions:

The isNaN function evaluates an argument to determine if it is "NaN" (not a number). The syntax of isNaN is:

`isNaN(testValue)`

where testValue is the value you want to evaluate.

The parseFloat and parseInt functions return "NaN" when they evaluate a value that is not a number. isNaN returns true if passed "NaN," and false otherwise.

The following code evaluates floatValue to determine if it is a number and then calls a procedure accordingly:

```
floatValue=parseFloat(toFloat)
if (isNaN(floatValue)){
    notFloat()
} else {
    isFloat()
}
```

Instructor Notes:

2.4: JavaScript Functions

Predefined Functions (Contd..)

➤

parseInt and parseFloat

- Returns a numeric value for string argument.

parseInt(str)
parseFloat(str)

parseInt(str, radix)
//returns an integer of specified radix of the string argument

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- 56 -

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parseInt and parseFloat Functions:

The two "parse" functions, `parseInt` and `parseFloat`, return a numeric value when given a string as an argument.

The syntax of `parseFloat` is:

`parseFloat(str)`

where `parseFloat` parses its argument, the string `str`, and attempts to return a floating-point number. If it encounters a character other than a sign (+ or -), a numeral (0-9), a decimal point, or an exponent, then it returns the value up to that point and ignores that character and all succeeding characters. If the first character cannot be converted to a number, it returns "NaN" (not a number).

The syntax of `parseInt` is:

`parseInt(str[, radix])`

where `parseInt` parses its first argument, the string `str`, and attempts to return an integer of the specified radix (base), indicated by the second, optional argument, `radix`. For example, a radix of ten indicates to convert to a decimal number, eight octal, sixteen hexadecimal, and so on. For radices above ten, the letters of the alphabet indicate numerals greater than nine. For example, for hexadecimal numbers (base 16), A through F are used.

If `parseInt` encounters a character that is not a numeral in the specified radix, it ignores it and all succeeding characters and returns the integer value parsed up to that point. If the first character cannot be converted to a number in the specified radix, it returns "NaN." The `parseInt` function truncates the string to integer values.

Instructor Notes:

2.4: JavaScript Functions

Predefined Functions (Contd..)

➤ **Number and string**

- Converts an object to a number or a string.

Number (objectReference)
String (objectReference)

```
today = new Date (430054663215)
now = String(today)
// returns "Thu Aug 18 04:37:43 GMT-0700 (PDT) 1983"
```

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**Number and String Functions:**

The Number and String functions let you convert an object to a number or a string.

The syntax of these functions is:

Number(*objRef*)

String(*objRef*)

where *objRef* is an object reference. The following example converts the Date object to a readable string.

```
D = new Date (430054663215)
```

```
// The following returns
```

```
// "Thu Aug 18 04:37:43 GMT-0700 (Pacific Daylight Time) 1983"
```

```
x = String(D)
```

Instructor Notes:

2.4: JavaScript Functions

Global and Local Variables

➤ Code Snippet for scope of variables

```
<script>
  var companyName="CAPGEMINI"
  function displayName(){
    var employeeName="Tom"
    document.write("Welcome to "+companyName+", "+employeeName)
  }
</script>
```

Global Variable

Local Variable

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Scope of Variables:

JavaScript supports two variable scopes:

Global variables

Local variables

The local variable applies only within a function and limits the scope of the variable to that function. To declare a local variable, the variable name must be preceded by var, as shown following:

```
var MaxValue=0;
```

Any variable declaration that is not within a function, is treated as a global variable. The syntax to declare a global variable is the same as that for local variable.

Instructor Notes:

2.4: JavaScript Functions

Global and Local Variables

- Variables that exist only inside a function are called **Local variables**
- The values of such **Local variables** cannot be changed by the main code or other functions
- Variables that exist throughout the script are called **Global variables**
- Their values can be changed anytime in the code and even by other functions

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- 59 -

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Using Global and Local Variables:

You can choose between local and Global variables by using the following guidelines:

If the value of a variable is meant to be used by any part of the program, both inside and outside functions, the variable should be declared outside any function. This has the effect of making it global and modifiable by any part of the program. The best place to declare global variables is in the <head> block of the HTML document.

If the variable is needed only within a particular function, the variable should be declared inside that function.

If you want the value of a variable to be modified only by the main script of a single function, but you need to use it in another function, pass the variable as an argument to that function. This has the effect of making a copy of the variable and assigning its value to the argument. As the function works and modifies its own copy of the variable, it will not effect the original. Argument variables are automatically declared as local to that function. Even if the argument has the same name as the variable being passed, making changes to it does not effect the variable that was passed. The only exception to this is objects. When an object is passed as an argument, it is passed by reference as opposed to being passed by value. Instead of making a copy of the object, the function uses the original object. Changes made to an object's properties within the function have an effect on the original object.

Instructor Notes:

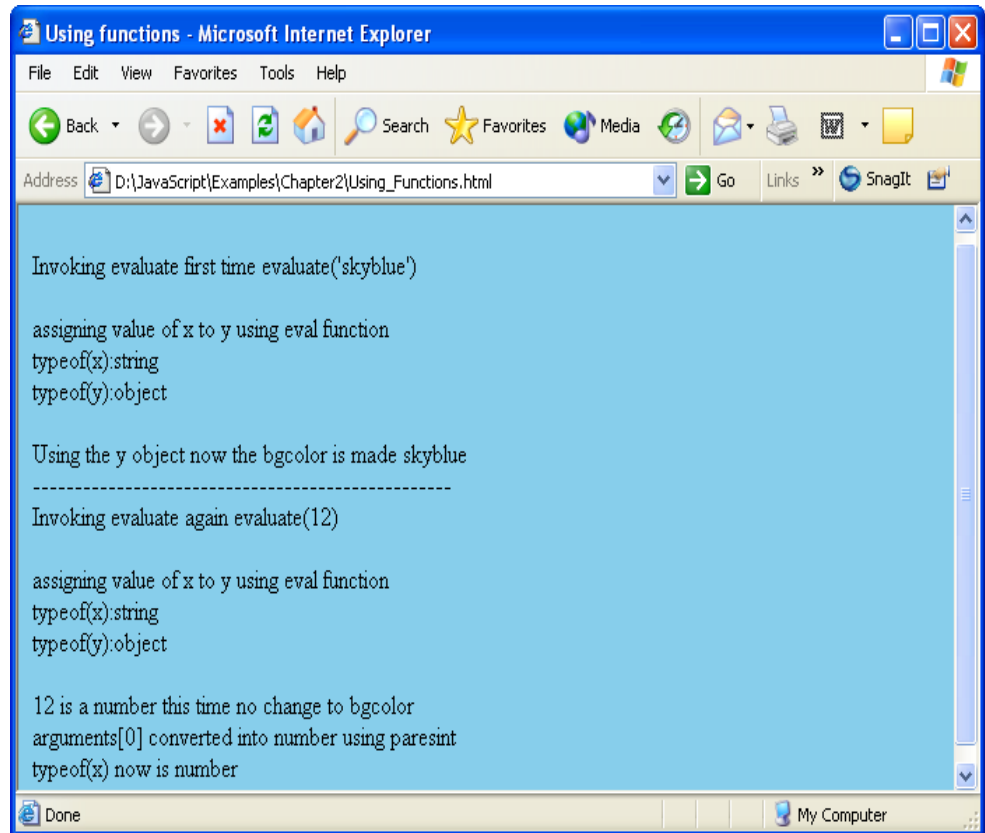
```

<HTML>
<HEAD>
<META NAME="GENERATOR" Content="Microsoft Visual Studio 6.0">
<TITLE>Using functions</TITLE>
<script language="Javascript">
<!--
function evaluate()
{
    var x;
    var y;
    x = "document";
    document.write("<br>assigning value of x to y using eval function");
    y = eval(x);
    document.write("<br>typeof(x):");
    document.write(typeof(x)+"<br>");
    document.write("typeof(y):");
    document.write(typeof(y)+"<br>");
    if(isNaN(arguments[0]))
    {
        document.write("<br>Using the y object now the bgcolor is made
                        "+arguments[0]);
        y.bgColor=arguments[0];
    }
    else
    {
        document.write("<br>"+arguments[0]+" is a number this time no
                        change to bgcolor");
        x=parseInt(arguments[0]);
        document.write("<br>arguments[0] converted into number using
                        parseInt ");
        document.write("<br>typeof(x) now is ");
        document.write(typeof(x)+"<br>");
    }
}
-->
</script></HEAD><BODY><script>
document.write("<br>Invoking evaluate first time evaluate('skyblue')<br>");
evaluate("skyblue");
document.write("<BR>");
for(var i=0;i<50;i++)
document.write(".");
document.write("<br>Invoking evaluate again evaluate(12)<br>");
evaluate(12);
</script>
</BODY>
</HTML>

```

Instructor Notes:

Example 2.5 Demo of creating functions and predefined function
(num_string_fun.html)
And it produces the output as:



Instructor Notes:

Demo

- If_else.html
- Switch_ex.html
- For_ex.html
- Break_con_ex.html
- Fun_ex.html
- Num_string_fun.html



Instructor Notes:

Lab

- **Lab 2 :**
- **The JavaScript language**



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- 63 -

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Instructor Notes:

Summary

- **Data Types & Variables**
 - Numbers, Strings, Boolean, and Null
- **Operators & Expressions**
- **Functions**
- **Predefined Functions**
 - eval, isFinite, isNaN, parseInt & parseFloat, Number & String
- **Global and Local variables**



Instructor Notes:

Answers

1. Option o
2. True

Review Question

➤ **Question 1: Which of the following two variable scopes is supported by JavaScript:**

- Global, Local
- Functional, Non functional
- Static, Dynamic



➤ **Question 2: The eval function evaluates a string of JavaScript code without reference to a particular object.**

- True/False

Instructor Notes:

Answers

- 1. 4
- 2. 5
- 3. 1
- 4. 2
- 5. 3

Review Question: Match the Following

1. Loop statements	1. isNaN
2. Arithmetic operators	2. +=, -=
3. Predefined function	3. &&,
4. Assignment operators	4. For, While
5. Logical operators	5. ++, --, %, *



Knowledge Check

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- 66 -



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Instructor Notes:



Web Basics - JavaScript

Lesson 3: Working with Predefined Core Objects

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Instructor Notes:

Lesson Objectives

- **Data Types in JavaScript**
- **String Object**
- **Math Object**
- **Date Object**



Instructor Notes:

3.1: Data Types in JavaScript

Data Types in JavaScript

➤ JavaScript has predefined objects and uses standard browser objects. Some of them are discussed here

➤ Predefined objects in JavaScript are:

- String
- Math
- Date

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- 69 -

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Instructor Notes:

3.2: String Object

String Object

- **Properties of a string object:**
 - Length: The length property returns the number of characters in a string.
 - Syntax : `stringObject.length`
 - `"Lincoln".length // result = 7`

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- 70 -

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String Objects: Creating a String

A string consists of one or more standard text characters between matching quote marks. JavaScript is forgiving in one regard: You can use single or double quotes, as long as you match two single quotes or two double quotes around a string.

JavaScript draws a fine line between a string value and a string object. Both let you use the same methods on their contents, so by and large, you do not have to create a string object (with the new `String()` constructor) every time you want to assign a string value to a variable. A simple assignment operation (`var myString = "fred"`) is all you need to create a string value that behaves on the surface very much like a full-fledged string object.

```
var myString = new String("characters")
var myString = "fred"
```

Properties of String Objects:**Length:**

The most frequently used property of a string is `length`. To derive the length of a string, extract its property as you would extract the `length` property of any object. The `length` value represents an integer count of the number of characters within the string. Spaces and punctuation symbols count as characters. Any backslash special characters embedded in a string count as one character, including such characters as newline and tab.

```
"Four score".length // result = 10
"One\ntwo".length // result = 7
"".length // result = 0
```

Instructor Notes:

3.2: String Object

String Object(Parsing Methods)

Pg

- **charAt(index):** The charAt() method returns the character at a specified position.
 - Syntax: String charAt(index)
 - "HelloWorld".charAt(5)// result "W"

- **concat()**
 - The concat() method is used to join two or more strings
 - One or more string objects to be joined to a string
 - Syntax: stringObject.concat(stringX,stringX,...,stringX)

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- 71 -

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String Objects (Parsing Methods):**String.charAt(index)**

Use the string.charAt() method to extract a single character from a string when you know the position of that character. For this method, you specify an index value in the string as a parameter to the method. The index value of the first character of the string is 0. If your script needs to get a range of characters, use the string.substring() method. It is a common mistake to use string.substring() to extract a character from inside a string, when the string.charAt() method is more efficient.

string.concat(string2)

Returns: Combined string.

"abc".concat("def")// result: "abcdef"

JavaScript's add-by-value operator (+) provides a convenient way to concatenate strings. N4 and IE4, however, introduces a string object method that performs the same task. The base string to which more text is appended is the object or value to the left of the period. The string to be appended is the parameter of the method, as the example demonstrates. Like the add-by-value operator, the concat() method doesn't know about word endings. You are responsible for including the necessary space between words if the two strings require a space between them in the result.

Instructor Notes:

3.2: String Object

String Object(Parsing Methods Contd)

➤ **match(regExpression)**

– Searches for a specified value in a string

– Syntax: string.match(regExpression)

➤ **replace(regExpression, replaceString)**

– Replaces some characters with some other characters in a string.

– Syntax: string.replace(regExpression, replaceString)

➤ **substr(start [, length])**

– Extracts a specified number of characters in a string, from a start index .

– Syntax: string.substr(start [, length])

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– 72 –

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String Objects (Parsing Methods contd..):

string.match(regExpression):Returns Array of matching strings.

The string.match() method relies on the RegExp (regular expression) object. The parameter must be a regular expression object. This method returns an array value when at least one match turns up; otherwise the returned value is null.

string.replace(regExpression,replaceString): Returns Changed string.

Regular expressions are commonly used to perform search-and-replace operations. JavaScript's string.replace() method provides a simple framework in which to perform this kind of operation on any string.

```
var str = "To be, or not to be: that is the question."
var regexp = /be/
str.replace(regexp, "exist")
```

string.substr(start [, length]) :Returns Characters of the string from the first character of the given string to the specified length.

Instructor Notes:

3.2: String Object

String Object(Converting Methods)

- **toLowerCase()**
 - Displays a string in lowercase letters
 - `string.toLowerCase()`
- **toUpperCase()**
 - Displays a string in uppercase letters
 - `string.toUpperCase()`

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String Objects (Parsing Methods contd..):

`string.toLowerCase()` and `string.toUpperCase()`

Returns: The string in all lower- or uppercase, depending on which method you invoke.

A great deal of what takes place on the Internet (and in JavaScript) is case-sensitive. URLs on some servers, for instance, are case-sensitive for directory names and filenames. These two methods, the simplest of the string methods, convert any string to either all lowercase or all uppercase. Any mixed-case strings get converted to a uniform case. If you want to compare user input from a field against some coded string without worrying about matching case, you should convert both strings to the same case for the comparison.

Instructor Notes:

3.2: String Object

String Object (Formatting Methods)

➤ **Formatting Methods:**

- `string.bold()` : Displays a string in bold
- `string.italics()` : Displays a string in italic
- `string.fontcolor (colorValue)` : Displays a string in a specified color
- `string.fontSize(integer1to7)` : Displays a string in a specified size
- `string.big()` : Displays a string in a big font
- `string.small()` : Displays a string in a small font

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- 74 -

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String Objects (Formatting Methods):

Now we come to the other group of string object methods, which ease the process of creating the numerous string display characteristics when you use JavaScript to assemble HTML code.

You can still use the standard HTML tags instead of by calling the string methods in your web pages. The choice is up to you. One advantage to the string methods is that they never forget the ending tag of a tag pair.

`string.fontSize()` and `string.fontcolor()` also affect the font characteristics of strings displayed in the HTML page. The parameters for these items are pretty straightforward—an integer between 1 and 7 corresponding to the seven browser font sizes and a color value (as either a hexadecimal triplet or color constant name) for the designated text.

Instructor Notes:

3.2: String Object

URL String Encoding and Decoding

- **JavaScript includes two functions for encoding & decoding**
 - escape()**
 - encodes the string that is contained in the string argument to make it portable.
 - So it can be transmitted across any network to any computer that supports ASCII characters.
 - unescape()**
 - Use the unescape function to decode an encoded sequence that was created using escape.

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- 75 -

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**URL String Encoding and Decoding**

```
escape("Howdy Pardner") // result = "Howdy%20Pardner"
```

```
unescape("Howdy%20Pardner") // result = "Howdy Pardner"
```

When browsers and servers communicate, some nonalphanumeric characters that we take for granted (such as a space) cannot make the journey in their native form. Only a narrower set of letters, numbers, and punctuation is allowed. To accommodate the rest, the characters must be encoded with a special symbol (%) and their hexadecimal ASCII values. For example, the space character is hex 20 (ASCII decimal 32). When encoded, it looks like %20. You may have seen this symbol in browser history lists or URLs.

JavaScript includes two functions, `escape()` and `unescape()`, that offer instant conversion of whole strings. To convert a plain string to one with these escape codes, use the `escape` function, as in `escape("Howdy Pardner") // result = "Howdy%20Pardner"`. The `unescape()` function converts the escape codes into human-readable form.

Instructor Notes:

Demo

- `string_len.html`
- `string_method.html`
- `string_style.html`



Instructor Notes:

3-3: Math Object Math Object - Properties & Methods	
Property/ Method	Description
Math.PI	PI (3.141592653589793116)
Math.SQRT2	Square root of 2 (1.4142)
Math.random()	Random number between 0 and 1
Math.round(val)	N+1 when $val \geq n.5$; otherwise N
Math.max(val1, val2)	The greater of $val1$ or $val2$
Math.min(val1, val2)	The lesser of $val1$ or $val2$

January 30, 2021

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- 77 -

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Math Object – Properties & Methods:

The Math object allows you to perform mathematical tasks. All properties and methods can be called without creating the Math object.

You can use them in your regular arithmetic expressions since they return constant values. For example, to obtain the circumference of a circle whose diameter is in variable d , you use the statement shown below.

$circumference = d * Math.PI$

The `Math.random()` method returns a floating-point value between 0 and 1. If you are designing a script to act like a card game, you need random integers between 1 and 52; for dice, the range is 1 to 6 per die. To generate a random integer between zero and any top value, use the first formula shown where n is the top number.

To generate random numbers between a different range use the second formula where m is the lowest possible integer value of the range and n equals the top number of the range minus m . In other words $n+m$ should add up to the highest number of the range you want. For the dice game, use the third formula for each throw of the die.

$Math.round(Math.random() * n)$

$Math.round(Math.random() * n) + m$

$newDieValue = Math.round(Math.random() * 5) + 1$

Instructor Notes:

Demo

> Rand_fun.html



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- 78 -

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Instructor Notes:

3.4: Date Object

Date

- **Properties and Methods:**
- `var dateObject = new Date([parameters])`

Properties	Description
<code>dateObj.getTime()</code>	Milliseconds since 1/1/70 00:00:00 GMT
<code>dateObj.getYear()</code>	Specified year minus 1900
<code>dateObj.getMonth()</code>	Month within the year (January = 0)
<code>dateObj.getDate()</code>	Date within the month

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- 79 -

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**Date Object: Creating a Date object**

```
var dateObject = new Date([parameters])
new Date("Month dd, yyyy hh:mm:ss")
new Date("Month dd, yyyy")
new Date(yy,mm,dd,hh,mm,ss)
new Date(yy,mm,dd)
new Date(milliseconds)
```

The Date object evaluates to an object rather than to some string or numeric value. If you leave the parameters empty, JavaScript takes that to mean you want today's date and the current time to be assigned to that new Date object. To create a Date object for a specific date or time, you have five ways to send values as a parameter to the new Date() constructor function.

The Date object has only a prototype property, which enables you to apply new properties and methods to every Date object created in the current page.

Instructor Notes:

3.4: Date Object

Date and Time Arithmetic

- To simplify the tasks of formatting and manipulating dates, JavaScript provides a Date object along with some extra functions that help you work with dates.

```
var oneMinute = 60 * 1000
var oneHour = oneMinute * 60
var oneDay = oneHour * 24
var oneWeek = oneDay * 7
targetDate = new Date()
dateInMs = targetDate.getTime()
dateInMs += oneWeek
targetDate.setTime(dateInMs)
```

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Date and time arithmetic:

You may need to perform some math with dates for any number of reasons. Perhaps you need to calculate a date at some fixed number of days or weeks in the future or figure out the number of days between two dates. When calculations of these types are required, remember the *lingua franca* of JavaScript date values: the milliseconds.

What you may need to do in your date-intensive scripts is establish some variable values representing the number of milliseconds for minutes, hours, days, or weeks, and then use those variables in your calculations. On the slide you can see an example that establishes some practical variable values, building on each other. With these values established in a script, you can use one to calculate the date one week from today. Following is the complete example.

Instructor Notes:

Date and time arithmetic (contd.):

```

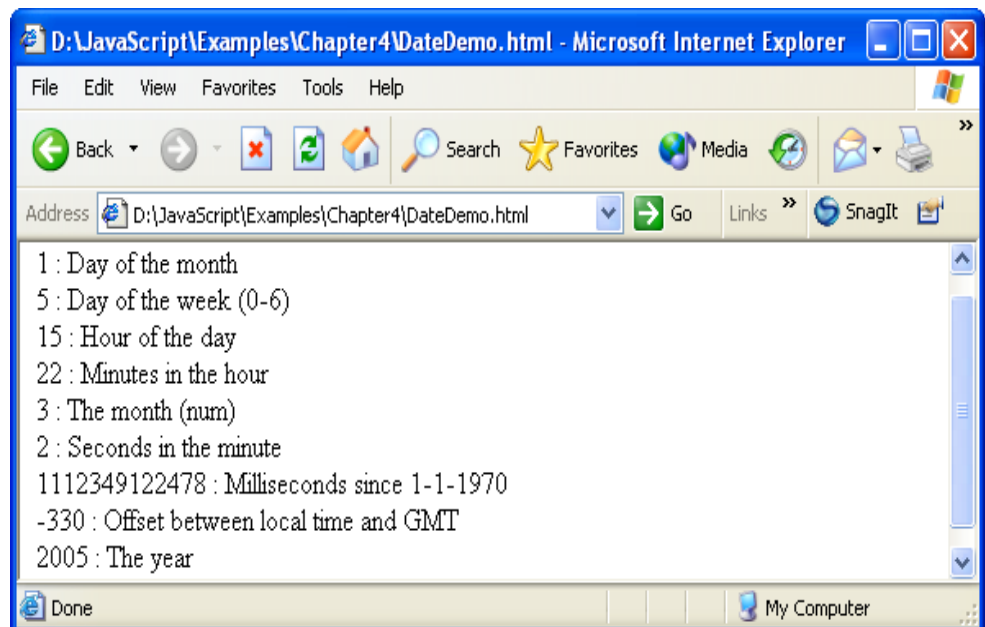
<!DOCTYPE html>
<HTML>
<HEAD>
<TITLE></TITLE>
</HEAD>
<BODY>
<script>
dateinfo = new Date();
document.write(dateinfo.getDate() + " : Day of the month" + "<br>")

document.write(dateinfo.getDay()           + " : Day of the week (0-6)"
+ "<br>")
document.write(dateinfo.getHours() + " : Hour of the day " + "<br>")

document.write(dateinfo.getMinutes() + " : Minutes in the hour" + "<br>")
document.write(dateinfo.getMonth() + " : The month          (num)" +
"<br>")
document.write(dateinfo.getSeconds() + " : Seconds in the minute " +
"<br>")
document.write(dateinfo.getTime() + " : Milliseconds since 1-1-1970" +
"<br>")
document.write(dateinfo.getTimezoneOffset() + " : Offset between local
time and GMT" + "<br>")
document.write(dateinfo.getFullYear() + " : The year" + "<br>")
</script>
</BODY>
</HTML>

```

And it produces the output as:



Instructor Notes:

Demo

- `Date_info.html`
- `DateDifference.html`
- `Utc_ex.html`



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- 82 -

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Add the notes here.

Instructor Notes:

Lab

- **Lab 3:**
- **Working with Predefined Core Objects**



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- 83 -

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Add the notes here.

Instructor Notes:

Summary

- **Predefined objects of JavaScript like String, Math and Date**
- **How to use predefined objects**
- **How to manipulate their properties and invoke methods**



Instructor Notes:

Answers:

Question 1: option 1

Question 2: False

Question 3: escape

Review Question

- **Question 1: Which is the method to extract a single character from a string when you know the position of that character.**
 - Option 1: `string.charAt()`
 - Option 2: `string.charAtIndex()`
- **Question 2: `getDate()` returns the day within the month.**
 - True/False
- **Question 3: To convert a plain string to one with these escape codes, use the _____ function.**



Instructor Notes:



Web Basics-JavaScript

Lesson 4: Working with Arrays

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Instructor Notes:

Lesson Objectives

- **In this lesson, you will learn about:**
 - Array object
 - Properties and methods of Array objects



Instructor Notes:

4.1: Array Object

Concept of Array Objects

➤ An array is the sole JavaScript data structure provided for storing and manipulating ordered collections of data

➤ For creating an array, you can use the following:

```
var solarSys = new Array() //empty array
```

```
solarSys = new Array(2) //Array defined with size  
solarSys[0] = "Mercury" // Assigning values to array  
solarSys[1] = "Venus"
```

```
solarSys = new Array("Mercury", "Venus") condensed array
```

```
solarsys=["Mercury",,, "Venus"] // literal array
```

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Array Objects:

An **array** is a kind of variable that can hold more than one value at a time.

However, unlike some other programming languages, JavaScript's arrays are very forgiving as to the kind of data you store in each cell or entry of the array. This allows, for example, an array of arrays, providing the equivalent of multidimensional arrays customized to the kind of data your application needs.

You can see a few examples listed on the slide for creating an array. We see example of creating an empty array. To limit the size of array you can specify the optional integer value as seen in the second example.

Another way of defining an array is called as condensed array which allows you to combine the array and array elements definitions into one step.

Literal arrays are define by assigning the value in square brackets. To create an array with initial undefined values you can simple enter a comma. For eg
myarray=["Pune",,, "Mumbai"]

Instructor Notes:

4.2: Properties and methods of Array Object

Concept of Array Object Methods

- **JavaScript provides the following array object methods:**
 - `arrayObject.length`
 - `arrayObject.join(separatorString)`
 - `arrayObject.reverse()`
- **Example for length method**
 - `myArray.length`// result: 5
- **Code snippet for usage of join method**
 - In this, myArray contents will be joined and placed into arrayText by using the comma separator

```
var arrayText = myArray.join(",")
```

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- 89 -

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Array Object Methods:

After you have information stored in an array, JavaScript provides several methods to help you manage that data.

`arrayObject.join(separatorString)`

It returns string of entries from the array delimited by the `separatorString` value.

```
var arrayText = myArray.join(",")
```

`arrayObject.reverse()`

It returns array of entries in the op.

The element that was last in the array becomes the 0 index item in the array. Note that when you do this, you are restructuring the original array, and not copying it.

Instructor Notes:

Demo

➤ Arr_demo.html



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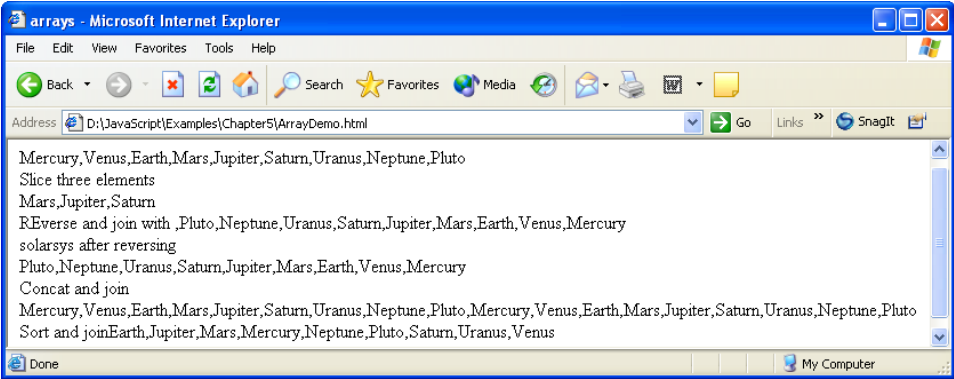
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- 90 -

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Demo of Array object (Arr_Demo.html) produces the output as shown below:



Instructor Notes:

Lab

- **Lab 4 :**
- **Working with Arrays**



Instructor Notes:

Summary

- An array is a set of variables
- You can create an array object by using new operator
- Array Object properties are length
- Array Object Methods are concat, join, reverse, slice etc



Instructor Notes:

Answers:

Question 1: option 2

Question 2: False

Review Question

- **Question 1: The ____ method allows you to join array contents and place it into a text**
 - Option 1: array.concat()
 - Option 2: array.join()
- **Question 2: An array object automatically has a size property.**
 - True/False



Instructor Notes:



Web Basics-JavaScript

Lesson 5: Document Object Model

January 30, 2021 Proprietary and Confidential - 94 -

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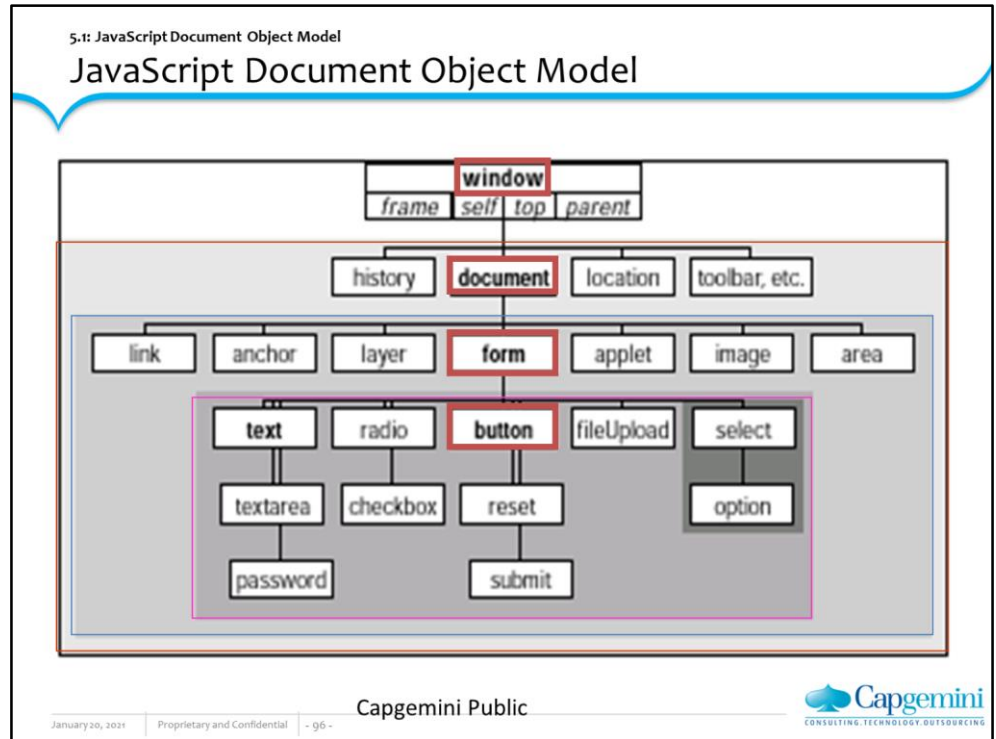
Instructor Notes:

Lesson Objectives

- **After completing this module you will be able to:**
 - Understand the JavaScript Object Model
 - Understand the Window object and Navigator Object
 - Working with Location and History Object

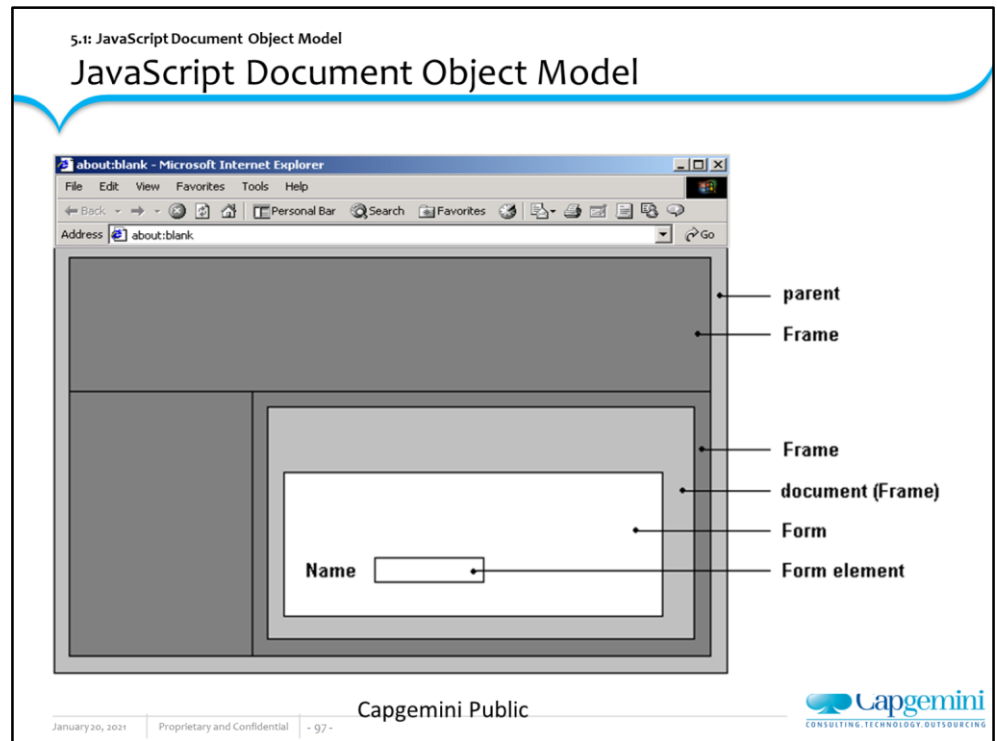


Instructor Notes:



The figure shows the complete JavaScript document object hierarchy as implemented in Netscape Navigator 4. Notice that the window object is the topmost object in the entire scheme. Everything you script in JavaScript is in the browser's window, be it the window itself or a form element. Of all the objects shown in the figure, you are likely to work most with the ones that appear in **boldface**. Objects whose names appear in *italics* are synonyms for the window object, and are used only in some circumstances. Pay attention to the shading of the concentric rectangles. Every object in the same shaded area is at the same level relative to the window object. When a link from an object extends to the next darker shaded rectangle, that object contains all the objects in darker areas. There exists at most one of these links between levels. A window object contains a document object; a document object contains a form object; a form object contains many different kinds of form elements. Study this figure to establish a mental model for the scriptable elements of a Web page. After you script these objects a few times, the object hierarchy will become second nature to you — even if you do not remember every detail (property, method, and event handler) of every object. At least you know where to look for information.

Instructor Notes:



Creating JavaScript Objects

Most of the objects that a browser creates for you are established when an HTML document loads into the browser. The same kind of HTML code you used to create links, anchors, and input elements tell a JavaScript-enhanced browser to create those objects in memory. The objects are there whether or not your scripts call them into action.

The only visible differences to the HTML code for defining those objects are one or more optional attributes specifically dedicated to JavaScript. By and large, these attributes specify the event you want the user interface element to react to and what JavaScript should do when the user takes that action. If you rely on the document's HTML code to perform the object generation, you spend more time figuring out how to do things with those objects or have them do things for you. Bear in mind that objects are created in their load order, which is why you should put most, if not all, deferred function definitions in the document's Head. If you create a multi-frame environment, a script in one frame cannot communicate with another frame's objects until both frames load.

Instructor Notes:

5.1: JavaScript Document Object Model

Object Properties

- Define a particular, current setting of an object
- Property names are case-sensitive
- Each property determines its own read-write status
- Any property you set survives as long as the document remains loaded in the window
- For example:

```
document.forms[0].phone.value = "555-1212"  
document.forms[0].phone.delimiter = "-"
```

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Object Properties

A property generally defines a particular, current setting of an object. The setting may reflect a visible attribute, such as a document's background color. It may also contain information that is not so obvious, such as the form *action* and *method* when it is submitted.

Document objects have most of their properties assigned by attribute settings of HTML tags that generate the objects. Thus, a property may be a string (for example, a name) or a number (for example, a size). A property can also be an array, such as an array of images contained by a document. If the HTML does not include all attributes, the browser usually provides default value for both attributes and corresponding JavaScript properties.

When used in script statements, property names are case-sensitive. Therefore, if you see a property name listed as *bgColor*, you must use it in a script statement with that exact case usage. But when you set an initial value of a property by way of an HTML attribute, the attribute name (like all of HTML) is not case-sensitive. Thus, `<BODY BGCOLOR="white">` and `<body bgcolor="white">` both set the same property value.

Object Methods

An object's method is a command that a script can give to that object. Some methods return values, but that is not a prerequisite for a method. Also, not every object has methods defined for it. In a majority of cases, invoking a method from a script causes some action to take place. It may be an obvious action, such as resizing a window, or something more subtle, such as processing a mouse click.

Instructor Notes:

5.1: JavaScript Document Object Model

Event Handlers

- Specify how an object reacts to an event
 - Event can be triggered by a user action or a browser action.
- There are two ways to map functions to events
 - Event handlers as methods:

```
document.formName.button1.onclick=f1()
```

- Event handlers as properties:

```
<INPUT TYPE="button" NAME="button1" onClick="f1()">
```

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- 99 -

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Object Event Handlers

Event handlers specify how an object reacts to an event, whether the event is triggered by a user action (for example, a button click) or a browser action (for example, the completion of a document load). Event Handlers can be specified as methods or they can be specified using attributes in tags.

Instructor Notes:

5.2: Window Object

Working with Window Object

- **Window object:**
 - Unique position at the top of the JavaScript object hierarchy
 - Can be omitted from object references since everything takes place in a window
 - The following two statements are the same
 - `window.alert("Welcome to Javascript ")`
 - `alert("Welcome to Javascript ")`
- **Properties**
 - `defaultStatus`
 - `status`
 - `closed`

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- 100 -

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About this Object

The `window` object has the unique position of being at the top of the JavaScript object hierarchy. This exalted location gives it a number of properties and behaviors unlike any other object. Among the list of properties for the window object is one called `self`. This property is synonymous to the window object itself. When you start your browser, it usually opens a window. That window is a valid window object, even if it is blank. This object is also the level at which a script asks the browser to display any of the three styles of the dialog boxes (a plain alert dialog box, an OK-Cancel confirmation dialog box, or a prompt for user text entry).

Property	Description
<code>defaultStatus</code>	<code>window.defaultStatus</code> property is normally an empty string, it sets or returns the default text which is in the statusbar of the window
<code>Status</code>	This property sets a text value to be displayed in the status bar
<code>closed</code>	Returns a boolean value which indicated if the window has been closed or no

Instructor Notes:

5.2: Window Object

Window Object Methods

- **alert(message)**

```
window.alert("Display Message")
```



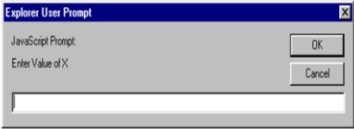
- **confirm(message)**

```
window.confirm("Exit Application ?")
```



- **prompt(message,[defaultReply])**

```
var input= window.prompt("Enter value of X")
```



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Method	Description
alert(message)	An alert dialog box is a modal window that presents a message to the user with a single OK button to dismiss the dialog box.
confirm(message)	A confirm dialog box presents a message in a modal dialog box along with OK and Cancel buttons. Such a dialog box can be used to ask a question of the user, usually prior to a script performing actions that will not be undoable.
prompt(message, defaultReply)	The third kind of dialog box that JavaScript can display includes a message from the script author, a field for user entry, and two buttons (OK and Cancel).

Instructor Notes:

5.2: Window Object

Window Object Methods

- `open("URL", "windowName" [, "windowFeatures"])`

```
newwin=window.open("new/URL","NewWindow","toolbar,status,resizable")
```

- `close()`
- `moveBy(deltaX,deltaY), moveTo(x,y)`
- `scrollBy(deltaX,deltaY), scrollTo(x,y)`

January 20, 2021 Proprietary and Confidential - 102 -

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<code>open("URL", "windowName" [, "windowFeatures"])</code>	The <code>window.open()</code> method, provides a Web site designer with options for the way a new browser window should look on the user's computerscreen. The optional <code>windowFeatures</code> parameter is <i>one string</i> , that comprises a comma-separated list of assignment expressions. Boolean values for true can be either yes, 1, or just the feature name by itself; for false, use a value of no or 0. If you omit any Boolean attributes, they are rendered as false. Therefore, if you want to create a new window that shows only the toolbar and statusbar and is resizable, the method looks like this: <code>window.open("newURL","NewWindow", "toolbar,status,resizable")</code> .
<code>close()</code>	The <code>window.close()</code> method closes the browser window referenced by the window object.
<code>scrollBy(deltaX,deltaY) scrollTo(x,y)</code>	<code>scrollBy(..)</code> method scrolls the content by the specified number of pixels which is relative scroll. <code>scrollTo(..)</code> is an absolute scroll to the specified coordinates
<code>moveBy(deltaX,deltaY) moveTo(x,y)</code>	<code>moveBy(..)</code> moves the window relative to the current position. <code>moveTo(..)</code> moves it to the specified coordinates

Instructor Notes:

5.2: Window Object

Window Object Methods

- **setTimeout, clearTimeout**

```
y=setTimeout('scroll()','100')
```

clearTimeout(y)

- **setInterval, clearInterval**

```
y=setInterval('scroll()','100')
```

clearInterval(y)

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setTimeout("functionOrExpr", msecDelay [, funcarg1, ..., funcargn])	Javascript holds a statement or function from executing for the desired amount of time. The timeout value is in milliseconds
setInterval("functionOrExpr", msecDelay, language)	Use this method when your script needs to call a function or execute some expression repeatedly with a fixed time delay between calls to that function or expression. The timeinterval is in milliseconds. Optional Language i.e Javascript, vbscript
clearInterval(intervalIDnumber)	Use this method to turn off an interval loop action started with the <code>window.setInterval()</code> method. The parameter is the ID number returned by the <code>setInterval()</code> method.
clearTimeout(timeoutIDnumber)	Use the <code>clearTimeout()</code> method in concert with the <code>window.setTimeout()</code> method when you want your script to cancel a timer that is waiting to run its expression. The parameter for this method is the ID number that the <code>setTimeout()</code> method returns when the timer starts ticking.

Instructor Notes:

5.2: Window Object

Window Object Event Handlers

- Event Handlers for the Window Object
 - onBlur
 - onFocus
 - onLoad

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- 104 -

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Event Handlers
Table 6.3 Window Object Event Handlers

Event Handler	Description
onBlur onFocus	Fired when window or frame has been activated and deactivated respectively.
onLoad	The load event is sent to the current window at the end of the document loading process.

Instructor Notes:

Demo

- Window_object.html
- setTimeout_method.html
- Window_ex.html
- setInterval_method.html



Instructor Notes:

5.3: Navigator Object

Navigator Object

- Netscape originally defined the navigator object for the Navigator 2 browser
- Microsoft Internet Explorer also supports the object in its object model
- The properties of the navigator object deal with the browser program the user runs to view documents

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- 106 -

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Navigator Object

Netscape originally defined the navigator object for the Navigator 2 browser. Microsoft Internet Explorer also supports the object in its object model. Properties of the navigator object deal with the browser program the user runs to view documents. Properties include those for extracting the version of the browser and the platform of the client running the browser.

Instructor Notes:

5.3: Navigator Object

Navigator Object

Properties	
appName	appCodeName
appVersion	userAgent
mimeTypes[]	Platform
plugins[]	cookieEnabled

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Property	Description
appName appCodeName appVersion userAgent Platform cookieEnabled	The <i>appName</i> and <i>appCodeName</i> properties are simply the official name and the internal code name for the browser. <i>appVersion</i> returns version information of the browser and <i>userAgent</i> returns the user-agent header sent by the browser to the server. Platform returns for which platform the browser is compiled. cookieEnabled determines if cookies are enabled in the browser
plugins[]	Returns an array of plugins available on the client browser.
mimeTypes[]	Returns an array of MIME types supported by the browser

Instructor Notes:

Demo

- `property_browser_name.html`
- `plugin_object.html`
- `MIME_Type.html`



Instructor Notes:

Lab

- Lab 5 :
- Working with Document Object Model (DOM)



Instructor Notes:

Summary

- Document Object Model is a interface that allows programs and scripts to dynamically access and update content, structure and style of documents
- Window object is the topmost object in the entire scheme. It has properties, methods and event handlers
- **The history object has an array of history items having details of the URL's visited from within that window**
- **The Location object contains information about the current URL**



Instructor Notes:

Answers

1. Option 2
2. True

Review Questions

- Question 1: Closed property returns _____ if the window object is closed either by a script or by the user.
 - Option 1: 1
 - Option 2: True
 - Option 3: 0
- Question 2: An alert dialog box is a modal window that presents a message for users with a single OK button to dismiss it.
 - True / False



Instructor Notes:

Answers

4. Option 2

5. Option 1

6. host

Review Questions (Contd..)

- Question 4: The _____ and appCodeName properties are simply the official name and the internal code name for the browser application.
 - Option 1: Appname
 - Option 2: appName
 - Option 3: appName
- Question 5: The _____ property supplies a string of the entire URL of the specified window object.
 - Option 1: location.href
 - Option 2: hostname
 - Option 3: hash
- Question 6: The _____ property describes both the hostname and port of a URL.



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Instructor Notes:



Web Basics-JavaScript

Lesson 6: Working With Document Object

January 20, 2021	Proprietary and Confidential	- 113 -	Capgemini Public	 CONSULTING TECHNOLOGY OUTSOURCING
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Instructor Notes:

Lesson Objectives

- **To understand the following topics:**
 - Document Object and its properties and methods
 - Cookies object



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- 114 -



Instructor Notes:

6.1: Document Object

Working With Document Object

- **Container for all HTML HEAD and BODY objects associated within tags**
- **Provides access to page elements from your script**
 - This includes form, link, anchor, as well as global Document properties such as background and foreground colors

January 26, 2021

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- 115 -

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Document object is part of the Window object. It is used to access all elements in a page. It provides access to the elements in an HTML page from within the script. This includes the properties of every form, link and anchor (and, where applicable, any sub-elements), as well as global document properties such as background and foreground colors.

Instructor Notes:

6.1: Document Object

Document Object Properties

- `alinkColor`, `vlinkColor`, `bgColor`, `fgColor`, `linkColor`
- `anchors[]`
- `applets[]`
- `forms[]`
- `links[]`
- `title`

January 20, 2021

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- 116 -

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Property	Description
<code>alinkColor</code> <code>vlinkColor</code> <code>bgColor</code> <code>fgColor</code> <code>linkColor</code>	Get and set the properties of document – activated link, visited link, background color, foreground color (text) and hyperlink color.
<code>anchors[]</code> , <code>forms[]</code> , <code>links[]</code>	These properties retrieve array of values respectively as present in the document object
<code>title</code>	Gets the title of the document which occurs between the TITLE tags.

Instructor Notes:

6.1: Document Object

Document Object Methods

- `write(), writeln()`
- `getElementsByTagName()`
- `getElementById()`
- `getElementsByName()`
- `getElementsByClassName()`

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Property	Description
<code>write("string1", ...)</code> <code>writeln("string1", ..)</code>	Both of these methods send text to a document for display in its window. The only difference between the two methods is that <code>document.writeln()</code> appends a carriage return to the end of the string it sends to the document (you must still write a <code>
</code> to insert a line break).
<code>getElementById("#para1")</code>	This method locates the element whose id has been passed. The text within this element can then be accessed using properties <code>innerHTML</code> or <code>innerText</code>
<code>getElementsByTagName("p")</code>	This method locates all the elements which match the tagname passed. Each element of this type of tag can then be accessed in an array like manner
<code>getElementsByName()</code>	This method locates all the elements which match the name passed. Same name to many elements is usually given for radio buttons.
<code>getElementsByClass()</code>	This method locates all the elements which match the class name passed.

Instructor Notes:

Demo

- `Link_Anchor_object.html`
- `Meta_information.html`
- `locate_element_by_id.html`
- `locate_elements_by_tagname.html`
- `locate_elements_by_name.html`
- `locate_element_by_class_name.html`



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- 118 -

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Instructor Notes:

6.2: Working with Cookies

Working with Cookies

- Text files that Web sites place in your computer to help your browsers remember specific information
- Used to store user preferences for content or personalized pages
- Following function sets cookie values (expiration date is optional):

```
function setCookie(name, value, expire) {  
    document.cookie = name + "=" + escape(value)  
    + ((expire == null) ? "" : ("; expires=" + expire.toGMTString()));  
}
```

January 20, 2021

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- 119 -

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Using Cookies

Cookies are a mechanism for storing persistent data on the client in a file called `cookies.txt`. Because HyperText Transport Protocol (HTTP) is a stateless protocol, cookies provide a way to maintain information between client requests. This section discusses basic uses of cookies and illustrates with a simple example.

Each cookie is a small item of information with an optional expiration date and is added to the cookie file in the following format:

`name=value;expires=expDate;`

Name is the name of the datum being stored, and *value* is its value. If *name* and *value* contain any semicolon, comma, or blank (space) characters, you must use the *escape* and *unescape* functions to encode and decode them respectively.

expDate is the expiration date, in GMT date format:

Wdy, DD-Mon-YY HH:MM:SS GMT

Although it is slightly different from this format, the date string returned by the *Date* method *toGMTString* can be used to set cookie expiration dates.

The expiration date is an optional parameter indicating how long to maintain the cookie. If *expDate* is not specified, the cookie expires when the user exits the current browser session. Browser maintains and retrieves a cookie only if its expiration date has not yet passed.

Instructor Notes:

Limitations

Cookies have these limitations:

300 total cookies in the cookie file.

4 Kbytes per cookie, for the sum of both the cookie's name and value.

20 cookies per server or domain (completely specified hosts and domains are treated as separate entities and have a 20-cookie limitation for each, not combined).

Cookies can be associated with one or more directories. If your files are all in one directory, then you need not worry about this. If your files are in multiple directories, you may need to use an additional path parameter for each cookie.

Using Cookies with JavaScript

The *document.cookie* property is a string that contains all names and values of Navigator cookies. Use this property to work with cookies in JavaScript.

Instructor Notes:

Demo

- Ch9_ex1.cookie.html
- Demo_Cookies.html



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- 121 -

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Instructor Notes:

Lab

- Lab 7:
- Working with Document object.



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- 122 -



Instructor Notes:

Summary

- JavaScript Document Object contains HTML elements contained in the <head> and <body> sections of a web page
- All the anchors are contained in anchor array.
- All the links are contained in link array
- Cookies are small text files stored on the site visitor's computer by their browser



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January 26, 2021

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- 123 -



Summary

In this chapter, you understood:

- DOM structure
- How to work with Document Object
- How to work with cookies

Instructor Notes:

Answers

1. Option 1
2. anchors[]

Review Questions

- **Question 1: The _____ is the container for all HTML HEAD and BODY objects.**
 - Option 1: Document
 - Option 2: Object
 - Option 3: Container
- **Question 2: _____ property in document object retrieves an indexed array of anchors in a document.**



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January 26, 2021

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- 124 -



Instructor Notes:



Web Basics-JavaScript

Lesson 7: Working with Form Object

January 20, 2021 Proprietary and Confidential - 125 - Capgemini Public



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Instructor Notes:

Lesson Objectives

- **To understand the following topics:**
 - Form Object Properties, Methods & Event Handlers
 - Text-Related Objects
 - Button Objects
 - Check Box and Radio Objects
 - Select Objects



January 20, 2021

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- 126 -

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Instructor Notes:

7.1: Form Object Properties, Methods and Event Handlers

Form Object

Properties	Methods	Event Handlers
action	reset()	onReset
elements[]	submit()	onSubmit
enctype		
length		
method		
name		
target		

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Working with Form Objects: Form Object Properties:

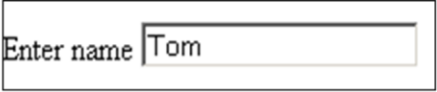
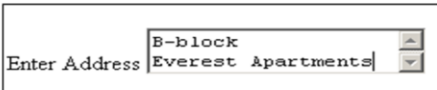
A form element provides the only way that users can enter textual information or make a selection from a predetermined set of choices, whether those choices appear in the form of an on/off checkbox, one of a set of mutually exclusive radio buttons, or a selection from a list.

Property/ Method/ Events	Description
action	This property is the same as the value you assign to the ACTION attribute of a <FORM> tag. The value is typically a URL on the server where queries or postings are sent for submission.
elements[]	Returns an array of elements. It includes all the user interface elements defined for a form: text fields, buttons, radio buttons, checkboxes, selection lists, and more.
encoding	You can define a form to alert a server that the data being submitted is in a MIME type. This property reflects the setting of the ENCTYPE attribute in the form definition. The default value is an empty string.
method	A form's method property is either the GET or POST values assigned to the METHOD attribute in a <FORM> tag.

Instructor Notes:

7.2: Text-Related Objects

Text-Related Objects

- **Text**
A text input field with the label "Enter name" and the text "Tom" entered.
- **Password**
A password input field with the label "Enter Password" and masked text "kkkkkkkkkk" entered.
- **TextArea**
A text area input field with the label "Enter Address" and the text "B-block Everest Apartments" entered.
- **Hidden Objects**

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Text-Related Objects:

Text Objects : The text object is the primary medium for capturing user-entered text.

Password Object: A password-style field looks like a text object, but when the user types something into the field, only asterisks or bullets (depending on your operating system) appears in the field.

Textarea Object: A textarea object closely resembles a text object, except for attributes that define its physical appearance on the page.

Hidden object: A hidden object is a simple string holder within a form object whose contents are not visible to the user of your Web page. With no methods or event handlers, the hidden object's value to your scripting is as a delivery vehicle for strings that your scripts need for reference values or other hard-wired data.

Instructor Notes:

7.2: Text-Related Objects Text-Related Objects (Contd..)		
Properties	Methods	Event Handlers
defaultValue	blur()	OnBlur
name	focus()	OnChange
type	select()	OnFocus
value		

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The properties, methods and event handlers are same for text object, text area and Password. For hidden object the properties are same but no methods and event handlers are associated with this object.

Property/ Events/ Methods	Description
defaultValue	Specifies or returns a defaultValue for a text related objects.
name	This property can be used to reference the text object in the script.
type	Returns the type of text related object
value	A reference to an object's value property returns the string currently displayed in the field.

Instructor Notes:

7.4: Check Box and Radio Objects

Check Box And Radio Objects

- **Checkbox**
- **Radio**

Properties	Methods	Event Handlers
checked	click()	OnClick
defaultChecked		
name		
type		
value		

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Checkbox object:

Property/ Events/ Methods	Description
checked	The simplest property of a checkbox gets or lets you set whether or not a checkbox is checked. The value is true for a checked box and false for an unchecked box. Only one radio button in a group can be highlighted checked) at a time. That one button's checked property is set to true, whereas all others in the group are set to false.
defaultChecked	If you add the CHECKED attribute to the <INPUT> definition for a checkbox or radio button, the defaultChecked property for that object is true; otherwise, false.
name	The name property allows user to access name for the checkbox or radio button through script.
type	Use the type property to help you identify a checkbox object or a radio button object from an unknown group of form elements.

Instructor Notes:

7.5: Select Objects

Select Object

SELECT

Properties	Methods	Event Handlers
length	blur()	onChange
name	focus()	onFocus
selectedIndex		onBlur
type		

OPTION

Default Selected
 text
 selected

Properties

January 20, 2021 Proprietary and Confidential - 131 -

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Property/ Methods/ Events	Description
length	Returns the number of items available in the list. A select object with three choices in it has a length property of 3.
Name	A select object's name property is the string you assign to the object by way of its NAME attribute in the object's <SELECT> tag which can be
selectedIndex	When a user clicks on a choice in a selection list, the selectedIndex property changes to a number corresponding to that item in the list.
type	Use the type property to help you identify a select object from an unknown group of form elements.
blur() focus()	Your scripts can bring focus to a select object by invoking the object's focus() method. To remove focus from an object, invoke its blur() method. These methods work identically with their counterparts in the text object.
onChange	As a user clicks on a new choice in a select object, the object receives a change event that can be captured by the onChange event handler.

Instructor Notes:

Using 'this' keyword

The 'this' keyword can be used to reference the object which called the function. It can be used within a function scope or global scope and it receives a different value in each scope. Depending on which object has called the function the value of 'this' will differ. The 'this' keyword always points to the object that is calling a particular method.

Consider the example given below:

The 'this' keyword is used in the showColor() function of an object. In this context, this is equal to car, making this code functionality equivalent to the following code snippet

So the reason for using 'this' is you never know what kind of variable names you will use to create an object. By using 'this' you are sure to invoke the correct function with the correct value. Also it allows you to use the same function any number of times. To understand this consider the following code:

`alert(this.color); //outputs "red"`
In the above snippet both oCar1 and oCar2 refer to the same function. The function gives the output according to the object which called the function.

```
var oCar = new Object;
oCar.color = "red";
oCar.showColor = function () {
    alert(oCar.color); //outputs "red"
};
```

```
function showColor() {
    alert(this.color);
}
var oCar1 = new Object;
oCar1.color = "red";
oCar1.showColor = showColor;
var oCar2 = new Object;
oCar2.color = "blue";
oCar2.showColor = showColor;
oCar1.showColor(); //outputs "red"
oCar2.showColor(); //outputs "blue"
```

Instructor Notes:

Demo

- `Form_Object.html`
- `Select_option.html`
- `Element_array.html`
- `Enctype.html`
- `Hidden_value.html`



Instructor Notes:

Lab

- **Lab 8:**
- **Working with Form Object**



Instructor Notes:

Summary

- **Form Object** corresponds to an HTML input form constructed with the **FORM** tag
- **Forms** have their own properties, objects, methods & events
- A form can be submitted by calling the JavaScript submit method or clicking the form submit button
- JavaScript can do entry-level validation & do it very easily



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Summary

This module provided an understanding of:

Form object and its components.

How to create form objects.

How to handle events.

How to validate data.

How to submit a form.

Instructor Notes:

Answers:

1. Option 1
2. True
3. click()

Review Questions

- **Question 1:** A form's _____ property is either the GET or POST values assigned to the METHOD attribute in a <FORM> definition.
 - Option 1: Method
 - Option 2: Class
 - Option 3: Object
- **Question 2:** The intention of the click() method is to enact, via a script, the physical act of clicking a radio button.
 - True / False
- **Question 3:** A button's _____ method should replicate, via scripting, the human action of clicking that button.



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